

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-01

Subject: Fluidized-bed roasting furnace, 10-16 t/h, oxygen-enriched blast.

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting-leaching-electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal (budgetary quotation) for the item above. Project site: an operating metallurgical site in north-west Russia.

Brief description of the item: Calcine hold-up in the furnace 139 t at any one time; shell of carbon steel + asbestos + fireclay; hearth of heat-resistant steel and concrete. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer's works) and DDP project site in north-west Russia terms (Incoterms 2020), stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from order / advance payment);
3. overall dimensions and weights of the equipment and of the largest shipping blocks (for logistics studies);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements.

Annex 1. Technical requirements

To RFQ No. OVE75-RFQ-01 — values follow the Customer's input data and terms of reference; Basic Engineering (FEED) stage, subject to refinement at the next project stage.

RFQ item	Fluidized-bed roasting furnace, 10-16 t/h, oxygen-enriched blast
Summary	Calcine hold-up in the furnace 139 t at any one time; shell of carbon steel + asbestos + fireclay; hearth of heat-resistant steel and concrete
Specific requirements	Reference specifically for sulphating roasting of copper feed in a fluidized bed (not for oxidizing roasting of zinc concentrates); confirmation of operation on blast

	containing 96 % O ₂
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Equipment covered by this request:

1. Fluidized-bed roasting furnace (FBR) Roasting

Type / model	Characteristics per Input Data, part 1, table 23
Duty	Oxidizing roasting of charge based on copper concentrate from the concentrator
Key parameters	Capacity 10–16 t/h
Quantity	1
Materials	Shell — carbon steel, asbestos sheet, fireclay brick; hearth — heat-resistant steel and concrete
Dimensions, hold-up	Calcine hold-up in the furnace at any one time — 139.0 t

2. Screw charge slinger Roasting

Duty	Feeding of the charge into the fluidized-bed furnace
Key parameters	up to 20 t/h
Quantity	1
Materials	Stainless heat-resistant steel
Dimensions, hold-up	Ø150, L1500 mm
Power	3.5 kW

3. Blast blower Roasting

Duty	Generation of the fluidized bed in the fluidized-bed furnace
Key parameters	Process air 6,655 Nm ³ /h (25 °C) / 12,612 Nm ³ /h (455.3 °C); pressure 0.09–0.12 MPa
Quantity	Per the Terms of Reference for Basic Engineering, clause 3.4

4. Throttle damper Roasting

Duty	Control of the gas duct of the fluidized-bed furnace
Quantity	1
Power	1.0 kW

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

预算性技术商务报价询价函 编号 OVE75-RFQ-01

事由：沸腾焙烧炉，10-16 t/h，富氧鼓风。

KVANT LLC（俄罗斯联邦）正在开展铜冶炼厂建设项目的基础设计（FEED）——“焙烧—浸出—电积”联合工艺，年产阴极铜 75,000 吨——现请贵司就上述项目提交预算性技术商务报价（预算报价）。项目场地：俄罗斯西北部一座在运营的冶金厂区。

设备简要说明：炉内焙砂一次持料量 139 t；炉壳采用碳钢 + 石棉 + 粘土耐火材料；炉底采用耐热钢和混凝土。技术要求见本询价函附件 1。

请在报价中包含：

1. 预算价格——分别按 EXW（制造商工厂）及 DDP 至俄罗斯西北部项目场地（Incoterms 2020）条件报价，注明币种及报价有效期；
2. 制造周期及预计交货期（自订单/预付款之日起按周计）；
3. 设备单机及最大运输单元的外形尺寸和重量（用于物流研究）；
4. 类似供货业绩表：产品、工况介质及参数、供货年份、国家；
5. 与工艺介质接触部件的材质，以及供货范围（仪表、备件、技术文件）；
6. 公用工程消耗及安装要求（如有资料）。

本询价为基础设计（FEED）阶段的预算性询价：不构成要约，双方均不因此承担义务；详细技术数据表将在项目下一阶段发出。

回复期限：请在收到本询价函后 3（三）周内将报价发送至 stepan@kvantpro.com。请确认收到本询价函。

保密条款。本询价函及随附技术资料仅供编制报价使用，属保密信息，未经 KVANT LLC 事先书面同意，不得向第三方披露、复制或发布。提交报价不为双方设立任何法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1：本询价项目的技术要求。

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-01 — 参数依据业主原始数据及设计任务书；基础设计（FEED）阶段，后续阶段可能进一步细化。

询价项目	沸腾焙烧炉，10-16 t/h，富氧鼓风
简要说明	炉内焙砂一次持料量 139 t；炉壳采用碳钢 + 石棉 + 粘土耐火材料；炉底采用耐热钢和混凝土
特殊要求	须具备铜物料沸腾（流态化）硫酸化焙烧的业绩（非锌精矿氧化焙烧）；确认可在含 96 % O ₂ 的鼓风条件下运行

本询价涵盖的设备清单：

1. 沸腾焙烧炉（流态化焙烧炉） 焙烧

型式／型号	技术特性见原始数据第 1 部分表 23
用途	以选矿厂铜精矿为基础的炉料氧化焙烧
主要参数	生产能力 10-16 t/h
数量	1

材质	炉壳——碳钢、石棉板、粘土耐火砖；炉底——耐热钢和混凝土
尺寸、持料量	炉内焙砂一次持料量——139.0 t

2. 螺旋抛料加料机 焙烧

用途	向沸腾焙烧炉给入炉料
主要参数	至 20 t/h
数量	1
材质	不锈钢耐热钢
尺寸、持料量	Ø150、L1500 mm
电耗	3.5 kW

3. 鼓风机 焙烧

用途	在沸腾焙烧炉内形成流化床
主要参数	工艺空气 6,655 Nm ³ /h (25 °C) / 12,612 Nm ³ /h (455.3 °C) ；压力 0.09-0.12 MPa
数量	依据基础设计任务书第 3.4 条

4. 节流闸板 焙烧

用途	沸腾焙烧炉烟气管路的调节
数量	1
电耗	1.0 kW

报价中须确认：工况介质、温度、压力、接触部件材质、生产能力、外形尺寸及重量、能耗、制造周期、交货条件。